Sampling Methods & Bias

ANSWER KEY

Section 1: Probability Sampling Designs

1 Which sampling method gives every possible group of n individuals an equal chance of being chosen?

a. **Simple Random Sample (SRS) ✓**

b. Stratified Random Sample

c. Cluster Sample

d. Systematic Sample

By definition, a Simple Random Sample (SRS) of size n guarantees that every individual and every subset of n individuals has the exact same probability of selection.

2 A university administrator divides students by class year (Freshman, Sophomore, Junior, Senior) and takes an SRS of 50 students from each group. What sampling technique was used?

a. **Stratified random sampling ✓**

b. Cluster sampling

c. Systematic sampling

d. Convenience sampling

Dividing the population into homogeneous strata (class years) and selecting a random sample from every stratum is stratified random sampling.

3 A city divides its territory into 40 voting precincts. Five precincts are randomly chosen, and every resident in those 5 precincts is surveyed. What sampling method is this?

a. **Cluster sampling ✓**

b. Stratified sampling

c. Simple random sampling

d. Multistage convenience sampling

Dividing the population into heterogeneous clusters (precincts), randomly choosing whole clusters, and surveying all members within selected clusters is cluster sampling.

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Section 2: Selection Bias & Undercoverage

4 A local radio station asks listeners to text their opinion on a proposed city tax. What primary source of bias affects this sample?

a. **Voluntary response bias ✓**

b. Undercoverage of rural citizens

c. Nonresponse bias

d. Measurement bias

Voluntary response samples suffer from bias because participants self-select, heavily over-representing individuals with strong negative or extreme viewpoints.

5 Conducting an opinion poll by calling registered landline telephone numbers will most likely suffer from:

a. **Undercoverage of younger adults who only use mobile phones ✓**

b. Response bias due to interviewer intimidation

c. Nonresponse bias caused by excessive mailings

d. Confounding of demographic variables

Undercoverage occurs when certain subgroups of the target population (such as mobile-only households) are systematically omitted from the sampling frame.

6 Can increasing the sample size from 100 to $10,000$ eliminate bias in a voluntary response survey?

a. **No, a larger biased sample simply estimates the biased quantity more precisely ✓**

b. Yes, large samples eliminate all forms of sampling bias by the Law of Large Numbers

c. Yes, provided $n \geq 1000$

d. Only if the population is normally distributed

Sample size reduces sampling variability (random error), but cannot fix systematic design flaws or bias.

Section 3: Nonresponse and Response Bias

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7 A survey mailed to 1,000 households receives 65 completed responses. What is the primary concern for validity?

a. **Nonresponse bias, because non-respondents may hold systematically different views ✓**

b. Undercoverage bias

c. Stratification bias

d. Question wording bias

Nonresponse bias arises when individuals chosen for the sample cannot be contacted or refuse to participate, leading to non-representative results.

8 A police officer in uniform asks high school students face-to-face if they have ever consumed alcohol. Which bias will most likely distort the findings?

a. **Response bias, as students are likely to lie to authority figures ✓**

b. Selection bias

c. Voluntary response bias

d. Cluster bias

Response bias occurs when the interviewer, phrasing, or social desirability influences respondents to provide inaccurate or untruthful answers.

9 Which survey question contains wording bias?

a. **'Given the catastrophic national debt, do you support reckless government spending increases?' ✓**

b. 'Do you support or oppose increasing funding for public transit?'

c. 'What is your favorite beverage?'

d. 'How many days per week do you exercise?'

Using emotionally loaded words ('catastrophic', 'reckless') introduces leading question/wording bias.

Section 4: Parameters vs. Statistics

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10 In statistical notation, which symbol represents a population parameter rather than a sample statistic?

a. μ ✓

b. $\bar{x}$

c. s

d. $\hat{p}$

μ represents the true population mean (a parameter). $\bar{x}, s, \hat{p}$ are sample statistics calculated from sample data.

11 A random sample of 250 voters found that 54% favor a bond measure. The number 54% is a:

a. **Statistic ($\hat{p}$)** ✓

b. Parameter (p)

c. Standard error

d. Margin of error

Any numerical summary computed directly from a sample is a statistic.

12 What is sampling variability?

a. **The natural variation of sample statistics from one random sample to another** ✓

b. Mistakes made by the surveyor during data collection

c. Differences caused by biased question wording

d. Variability that occurs when sample size is too large

Sampling variability refers to the fact that different random samples from the same population will yield slightly different values of the statistic.

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